

SolarShield AI — Team Working Document

Problem Statement 15: Forecasting and/or Nowcasting of Solar Flares using combined Soft and Hard X-ray data from Aditya-L1 **Team size:** 2 (AI/ML + CSE) **Time remaining:** < 1 day **Status:** Build-mode. Everything below is scoped to what we can actually finish.

1. The One-Sentence Pitch

We combine Aditya-L1's SoLEXS (soft X-ray) and HELIOS (hard X-ray) data to detect solar flares as they happen and forecast them minutes ahead — using the physical fact that hard X-ray bursts precede soft X-ray thermal peaks — and we present this through an explainable dashboard, not a black-box probability number.

If you remember nothing else, remember this sentence. It's what you say when a judge stops you mid-demo.

2. What We Are Actually Building Tonight (Core Scope)

This is the **only** thing that needs to work end-to-end. Everything fancier is "Phase 2 / Roadmap" — mentioned in the deck, not promised as built.

#	Component	Owner	Output
1	Data fusion	AI/ML	One merged time-series dataframe: timestamp, soft flux, hard flux
2	Behavioural feature engineering	AI/ML	Rolling mean, rate-of-change, HXR/SXR ratio, HXR→SXR lag
3	Flare labeling	AI/ML	Binary flare/no-flare labels (from GOES catalogue or threshold rule)
4	Forecasting model	AI/ML	Simple classifier (XGBoost/logistic regression) outputting flare probability + estimated lead time
5	Solar Health Index	CSE + AI/ML	Rule-based state: Quiet / Watch / Warning / Critical, derived from model output + thresholds
6	Dashboard	CSE	Live-updating chart (soft + hard X-ray), health index banner, alert when flare nowcasted/forecasted
7	Deck	Both	8 slides, Core vs Future clearly separated

That's it. Seven things. If all seven exist in some working form, we have a demo-able product.

Explicitly NOT building tonight (these stay in the vision doc as roadmap, said out loud in the pitch as future work):

- Behaviour embeddings / vector similarity search
- Historical similarity retrieval ("89% similar to past M-class flare")
- Consensus engine combining multiple models
- 6-stage event evolution model
- Multi-payload expansion (VELC, SUIT, MAG, etc.)

Why we're cutting these: each one is a multi-week research component on its own. Claiming them as built and failing to demo them live is a worse outcome than not claiming them at all. A judge respects "here's what we built, here's where we're taking it" far more than a demo that breaks under questioning.

3. The Physics We're Leaning On (so everyone can explain it)

- **SoLEXS** = Solar Low Energy X-ray Spectrometer → captures **soft X-rays**, which track the slow thermal buildup of plasma during a flare.
- **HEL1OS** = High Energy L1 Orbiting X-ray Spectrometer → captures **hard X-rays**, produced almost instantly when accelerated electrons hit the lower solar atmosphere.
- **The key fact (Neupert effect):** the hard X-ray burst typically rises and peaks *before* the soft X-ray thermal peak. The soft X-ray channel is essentially the time-integral of the hard X-ray energy deposit.
- **Why this matters for us:** if we detect a sharp rise in hard X-ray flux, we can estimate — based on historical lag — how many minutes until the soft X-ray (and the visible "flare") peaks. That lag is our lead time. It's not a black box guessing; it's a measurable physical relationship.

Practice saying this in one breath. It's your strongest answer to "what's novel here?"

4. Data Pipeline (AI/ML)

4.1 Inputs

- SoLEXS Level-1 data (ISSDC PRADAN portal)
- HEL1OS Level-1 data (ISSDC PRADAN portal)
- Optional: GOES X-ray flare catalogue for ground-truth labels, if time allows. If not,

use a simple threshold rule (flux above $N\times$ background for M minutes = flare) for labeling.

4.2 Steps

1. Load both datasets, parse timestamps to a common format.
2. Resample/align both streams onto one shared timeline (same time resolution — pick whichever is coarser).
3. Merge into a single dataframe: `(timestamp | soft_flux | hard_flux)`.
4. Handle gaps/missing data — forward-fill short gaps, drop long ones.

4.3 Features to compute

Feature	What it captures
Rolling mean (soft, hard)	Smoothed baseline
Rate of change (soft, hard)	How fast flux is rising — precursor signal
HXR/SXR ratio	Relative dominance of hard vs soft emission
Cross-correlation lag (HXR→SXR)	The Neupert-effect lead time itself
Peak ratio	Hard peak height vs soft peak height
Event duration	How long the flare lasted, for context

4.4 Labeling

- Binary label: flare / no-flare at each timestamp (or per-window).
- If using GOES catalogue: match flare event times to our timeline and label $\pm$ window around each event.
- If no time for GOES matching: define flare = soft flux crosses a fixed threshold (e.g., consistent with C-class or above) for at least N consecutive minutes.

4.5 Model

- Start with the simplest thing that works: **logistic regression or XGBoost** on the feature table.
- Target: probability of flare in next N minutes (pick N = 10 as a reasonable, defensible number).
- Output the model needs to produce per timestamp:

- `flare_probability` (0-1)
- `predicted_class` (optional — e.g., low/high, or A/B/C/M/X if time allows)
- `estimated_lead_time_minutes` (derived from the HXR→SXR lag feature, not a separate model)

Do not attempt LSTM unless the simple model is already working and there's spare time. A working simple model beats a half-trained complex one every time.

5. Solar Health Index (shared logic, simple rules)

Map model output + thresholds into four states. This becomes the headline visual on the dashboard.

State	Condition (example)
Quiet	$\text{flare_probability} < 0.2$
Watch	$0.2 \leq \text{flare_probability} < 0.5$
Warning	$0.5 \leq \text{flare_probability} < 0.8$
Critical	$\text{flare_probability} \geq 0.8$, or active flare detected

This is a rule-based wrapper around the model output — not a separate ML system. Keep it simple; it just needs to look and feel meaningful on screen.

6. Dashboard (CSE)

6.1 Stack

Streamlit (fastest to build and demo). Avoid React/full-stack unless there's already a working skeleton — speed matters more than polish tonight.

6.2 Required elements

1. **Two live/scrolling charts:** soft X-ray flux, hard X-ray flux, time-aligned.
2. **Solar Health Index banner:** big colored label — Quiet (green) / Watch (yellow) / Warning (orange) / Critical (red).
3. **Alert box:** triggers visually when a flare is nowcasted (detected) or forecasted (predicted ahead of peak), showing estimated lead time.
4. **Confidence/probability readout:** simple number or bar.
5. (Optional, only if time allows) Predicted flare class badge (e.g., "M-class likely").

6.3 Data flow

feature dataframe → model output (probability, lead time) → health index

logic → Streamlit display

Keep the dashboard reading from a static/replayed dataset if live streaming isn't feasible — a "replay mode" that steps through historical data convincingly simulating real-time is a completely acceptable and common hackathon shortcut. Be upfront about this if asked; it's not deceptive, it's standard practice.

7. Roles & Division (2-person team)

AI/ML (you):

- Data loading, merging, feature engineering
- Labeling
- Model training and evaluation
- Producing the lead-time number and accuracy metrics for the deck

CSE (teammate):

- Streamlit dashboard build
- Wiring model output into the dashboard
- Repo structure, running the demo smoothly
- Helping assemble/format the deck once content exists

Both, last hour:

- Dry run of the demo, end to end
 - Final deck polish
 - Decide what to say if something breaks live (always have a screenshot/recording backup)
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8. Deck Structure (8 slides)

1. **Title** — SolarShield AI, problem statement, team
2. **Problem & Impact** — why flare forecasting matters (satellites, GPS, power grids)
3. **Our Insight** — the Neupert-effect framing: hard X-ray precedes soft X-ray, gives real lead time, physics-grounded not black-box
4. **System Architecture** — data fusion → features → model → health index →

dashboard (keep it to what's actually built)

5. **Results** — one chart: HXR rise vs SXR peak with lag annotated; accuracy/precision/recall numbers; lead-time number front and center
 6. **Dashboard Demo** — screenshot or live demo
 7. **What's Next (Roadmap)** — explicitly labeled as future work: behavioural embeddings, historical similarity retrieval, consensus engine, multi-payload expansion (VELC, SUIT, MAG). This is where the bigger vision lives — presented honestly as direction, not as built.
 8. **Team & Thank You**
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9. Key Numbers to Have Ready

Before the demo, make sure you can say, out loud, without checking notes:

- What's our detection accuracy / precision / recall?
- What's our median lead time (minutes before peak)?
- How many flare events are in our evaluation set?
- What does "Watch" vs "Warning" actually mean in plain English?

If any of these are fuzzy, that's the next thing to nail down — not new features.

10. Definition of Done (tonight)

We are demo-ready when:

- ☐ Merged dataframe with computed features exists
- ☐ Model produces flare probability + lead time on held-out data
- ☐ Health index logic converts that into Quiet/Watch/Warning/Critical
- ☐ Dashboard displays both charts + health banner + alert, fed by model output
- ☐ One results chart showing HXR→SXR lag is ready for the deck
- ☐ Deck has all 8 slides, Core vs Future clearly separated
- ☐ Backup screen recording of the dashboard exists
- ☐ Both people can explain the Neupert-effect pitch in one breath

That's the whole job. Build top to bottom, and if time runs out, cut from the bottom of section 2's table, not the top.